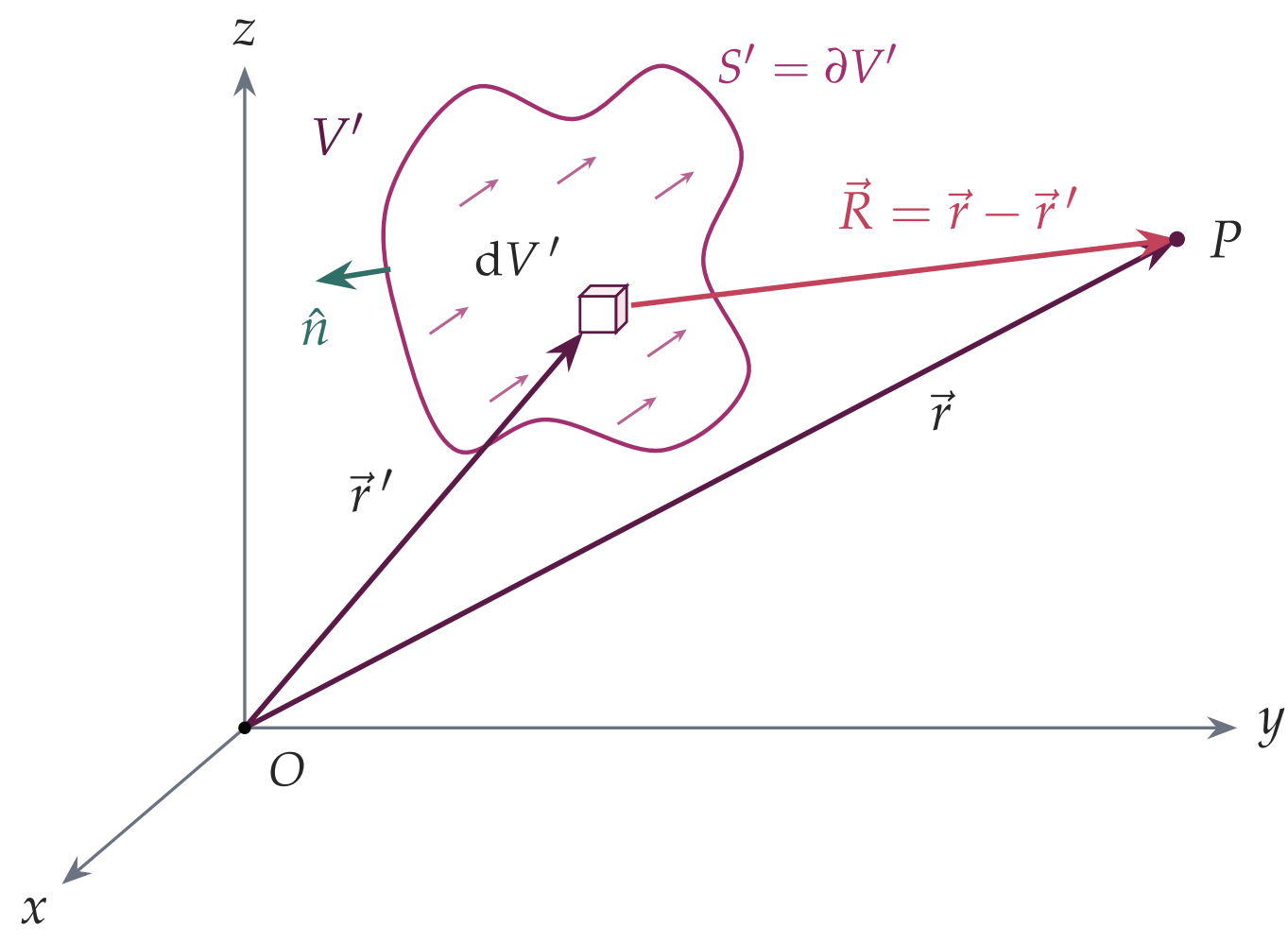


$$\mathrm{d}\vec{p} = \vec{P}(\vec{r}\,')\,\mathrm{d}V'$$

$$\mathrm{d}\phi(\vec{r}) = \frac{\mathrm{d}\vec{p}\cdot\hat{u}_R}{4\pi\varepsilon_0 R^2}$$

$$\rho_P = -\vec{\nabla}'\cdot\vec{P}, \quad \sigma_P = \vec{P}\cdot\hat{n}$$